

Element F

In our consideration of the viability of the originally selected design that connected the boxes via latches, we determined the weakest point of the system to be the points of connection by intuition. As such, we decided to analyze the strain in the latches at this point to mathematically measure the viability of this solution from a physics standpoint. We will begin from a simplified viewpoint of the problem, where we can assume that the boxes at the end are held up by a force acting on their center of gravity (to eliminate the necessity to worry about torque) because this will naturally produce less strain on the joints. If we calculate an undesired strain using this approach, there too would also be an undesired strain if we begin to worry about torque. If not, however, we should more accurately model our analysis to the problem at hand.

Given the equation for shear stress $\tau = \frac{dV}{dA}$, we may approximate the stress on a particular object if we take the shear force (V) to be constant. This provides the following derivation:

$$\tau = \frac{dV}{dA}$$

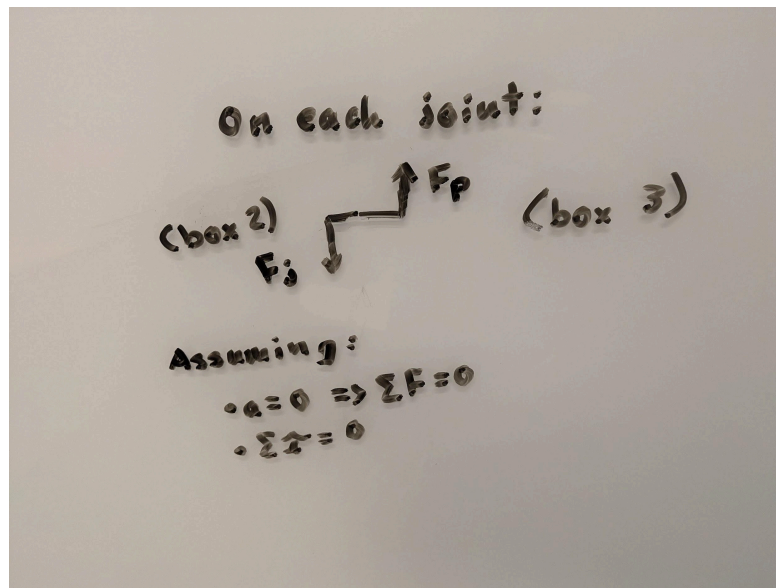
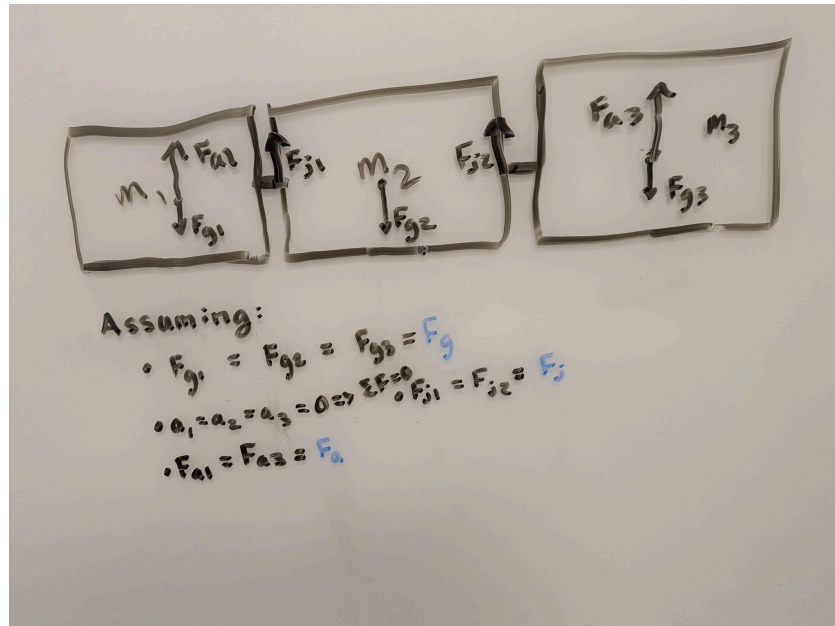
$$\tau dA = dV$$

$$\int \tau dA = \int dV$$

$$\tau_{avg} \int dA = V$$

$$\tau_{avg} = \frac{V}{A}$$

Following this equation, we now know that we must find the shear force acting on the latch (V) and the cross-sectional area upon which the force acts. To begin, we will determine the force that acts on the connection system between two boxes, drawing a free body diagram first to make the subsequent derivations more intuitive.



We can now further analyze only the middle box under static conditions to determine what the shear force is on the joint:

$$\Sigma F_2 = m_2 a = |F_{j1}| + |F_{j2}| - |F_g|$$

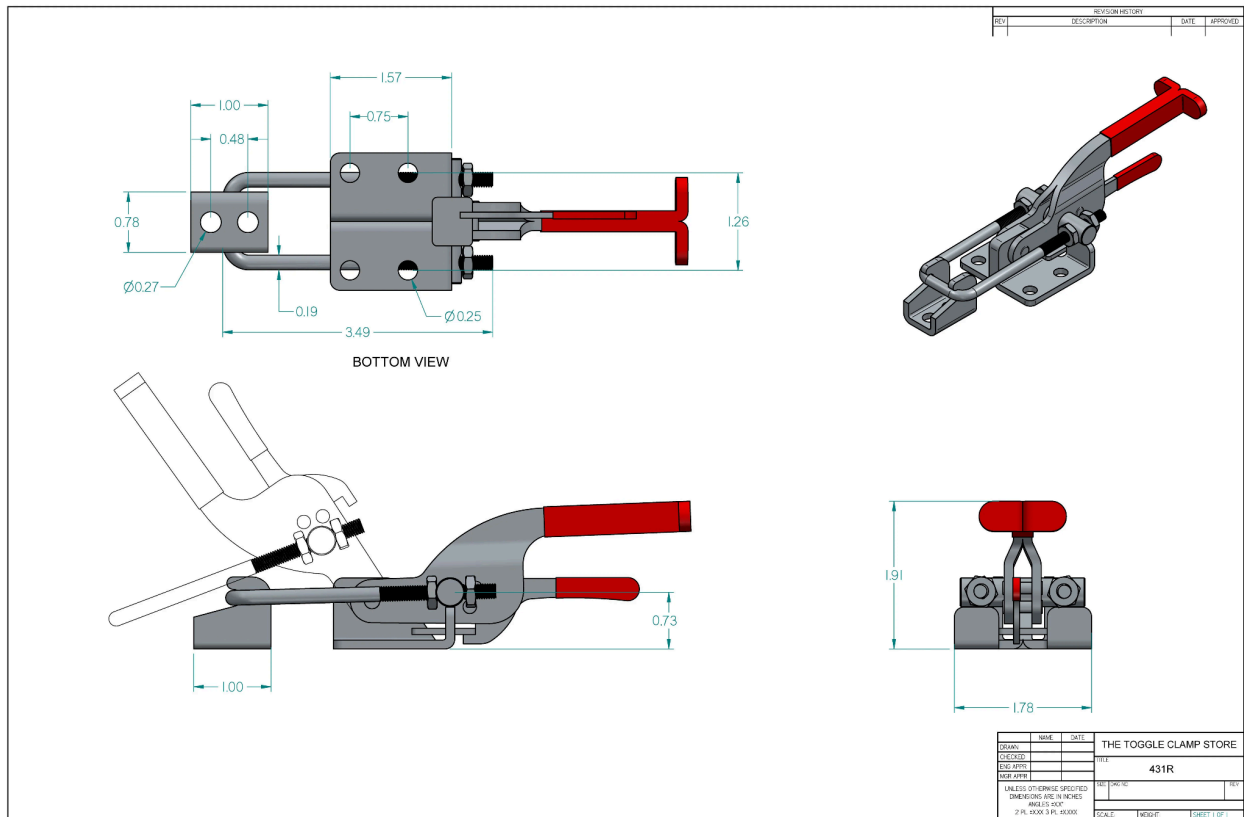
$$m_2(0) = 2|F_j| - |F_g|$$

$$0 = 2|F_j| - m_2g$$

$$|F_j| = \frac{m_2g}{2}$$

Thus we know $V = \frac{m_2g}{2}$, and we are now left only to determine the cross-sectional area of the joint that connects the boxes. Since the latch that we are discussing has two parallel bars joined at the end with minimal slack between the two (a similar version is shown below), only one of the bars will be in contact with the plate they attach to when a load is applied perpendicular to the bars on the plane that connects the two bars. That means that $A = n\pi r^2 = 0.000004\pi n$, where n is the number of latches.

Substituting this into the equation from earlier, we find $\tau_{avg} = \frac{m_2g}{0.000008\pi n^2}$



Now to find the angle of deflection we can expect in the system, we can use Hooke's Law for shear: $\gamma = \frac{\tau}{G}$, where $\gamma = \frac{\Delta x}{L} = \tan(\theta)$ and G is the shear modulus (79.3 GPa for steel), meaning we can rewrite this equation as follows to solve for the angle of displacement:

$$\gamma = \tan(\theta) = \frac{\tau}{G}$$

$$\theta = \tan^{-1}\left(\frac{\tau}{(79.3)(10^9)}\right)$$

$$\theta = \tan^{-1}\left(\frac{m_z g}{634400\pi n^2}\right)$$

Now, to solve for the number of latches needed to keep the angle of displacement under 5 degrees (arbitrarily chosen for example although we will reduce this to 0.001 degrees later), we can rewrite the equation as follows:

$$\tan\left(5 * \frac{\pi}{180}\right) = \frac{m_z g}{634400\pi n^2}$$

$$634400\pi n^2 = \frac{m_z g}{\tan\left(5 * \frac{\pi}{180}\right)}$$

$$n = \sqrt{\frac{m_z g}{(634400)(\pi) \tan\left(5 * \frac{\pi}{180}\right)}}$$

Substituting known values leaves us with:

$$n = \sqrt{\frac{(33.3)(9.81)}{(634400)(\pi) \tan\left(5 * \frac{\pi}{180}\right)}} \approx 0.0432 \text{ Latches}$$

This tells us that surprisingly if we are able to keep the torque at a manageable level, the boxes shouldn't deform too much (in fact, only 4 latches are needed to ensure the angle stays under 0.001 degrees, which would be more ideal). These calculations were checked by our engineering teacher and he confirmed them to be accurate to his knowledge. To ensure that the users don't have to latch a great number of latches and the torque is properly handled, we thus determined to use a modified solution which put the brunt of the weight on PVC pipes, which could be reinforced with steel thicker than the width of 4 latches if needed, and used the latches at the bottom to handle the torque, which would put the brunt of the force causing the torque against the largest cross-sectional area in the latch, thus reducing the stress on the latch. This also proved fiscally more feasible, which ultimately led us to conclude that the design for the connection system was entirely feasible.

Within the context of our problem, this will certainly do to hold the boxes together with negligible displacement while allowing the boxes to be picked up and moved around (while the strain will inevitably increase when picked up, it could not conceivably exceed 0.01 degrees, which is considered perfectly negligible by our standards). In terms of fiscal feasibility, the solution uses primarily resources that could be found on hand, and is thus extremely feasible (costing less than \$50). In terms of user experience, the design could be better, but we estimate that the boxes could be assembled and disassembled in under a minute, which means that it is certainly feasible to use in a real-life scenario although a more robust and quicker solution would be desirable.

As for the door, we decided to use tracks welded onto the box, which would definitely stand up to the force that could be conceivably exerted on them. We decided to continue with this solution since it was given to us as a constraint by the primary shareholder.